

BIG DATA ARCHITETURES

MESTRADO EM ENGENHARIA INFORMÁTICA – ENGENHARIA DE DADOS
ANO LETIVO: 2024/2025

ANA MADUREIRA

69

BIG DATA ARCHITECTURES

- **Data Architecture** is a strategic enabler for data and analytics success
- To effectively connect business strategy with technical execution.
- To plan effectively, connect strategy to actions and create long-term value for the organization.

The Role of Architecture Within the D&A Strategy and Operating Model



Source: Gartner
© 2023 Gartner, Inc. and/or its affiliates. All rights reserved. CM_073_2442296

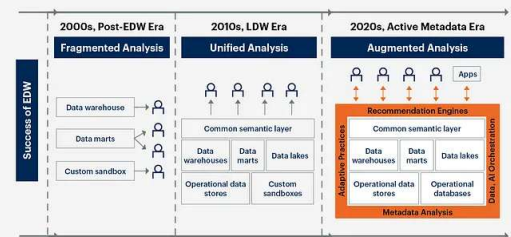
Gartner.

70

BIG DATA ARCHITECTURES

- Data architecture is evolving to deliver metadata-enabled data self-service.
- refers to the structured framework that outlines how data is collected, stored, managed, and used within an organization.
- it defines the data flows, the data models, and the technologies used to handle data.
- ensures that data is accessible, reliable, and relevant for various business needs, facilitating effective decision-making and operational efficiency.

The Evolution of Data Architecture



EDW = enterprise data warehouse
LDW = logical data warehouse
Source: Data Fabric or Data Mesh: How to Decide Your Future Data Management Architecture
© 2023 Gartner, Inc. and/or its affiliates. All rights reserved. Z98003

Gartner

71

BIG DATA ARCHITECTURES

Big data architecture is specifically designed to manage data ingestion, data processing, and analysis of data that is too large or complex. The different types of big data architecture frameworks depend on business's needs and goals:

- **Batch architecture** - Handles data in a batch
- **Lambda and Kappa architectures** - Methods for handling streaming data as it arrives
- **Streaming architectures** - Handles data in almost real-time
- **Hybrid architectures** - Uses both batch and streaming processes

72

TYPES OF BIG DATA ARCHITECTURE

- **Lambda Architecture** - handles both batch (static) data and real-time processing data. Uses separate batch and stream processing systems, making it scalable and fault-tolerant but complex to set up and maintain. It is well-suited when organizations have mixed requirements for stream and batch processing, e.g., for real-time analytics and multiple batch processing tasks or data lakes.
- **Kappa Architecture** - to handle both real-time streaming and batch processing data. Simplifies the pipeline with a single stream processing system as it treats all data as streams, providing flexibility and ease of maintenance. It is ideal for continuous data pipelines, real-time data processing, and IoT systems.

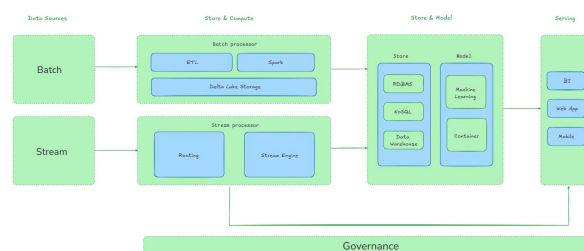
73

BIG DATA ARCHITECTURE - LAMBDA ARCHITECTURE

The Lambda Architecture is a data processing architecture that provides a solution for handling large volumes of data in a fault-tolerant and scalable manner. It was developed by Nathan Marz in 2011 as a solution to the challenges of processing data in real time at scale. The architecture is designed to process batch and real-time data streams simultaneously. It consists of three layers:

- **Batch layer** - is responsible for storing and processing all of the data in its raw, unprocessed form. It acts as the master dataset, which can be processed by the other two layers.
- **Speed layer** - processes real-time data streams as they arrive. It processes the data in real-time, and its output is fed into the serving layer. The speed layer typically uses a stream processing framework like Apache Kafka, Apache Storm, or Apache Samza for processing real-time data streams.
- **Serving layer** - provides low-latency access to the results of both the batch and speed layers. It serves as the interface between the processed data and the end-users who need to access it.

Lambda Architecture

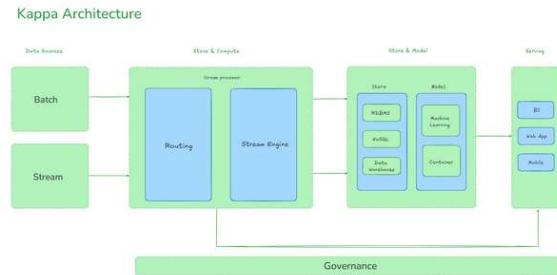


<https://www.chaosgenius.io/blog/lambda-architecture/>

74

BIG DATA ARCHITECTURE - KAPPA ARCHITECTURE

- The Kappa Architecture is a variation of the Lambda Architecture, which is designed to handle real-time data processing in a more streamlined and simplified way. It was proposed by Jay Kreps, one of the co-founders of Apache Kafka, in 2014:
 - has only one layer for processing real-time data streams, which is the same as the speed layer in the Lambda Architecture. It uses a stream processing system like Apache Kafka Streams, Apache Flink, or Apache Samza to process the data in real-time and store it in a database like Apache Cassandra or Apache HBase.
 - is simpler than the Lambda Architecture because it eliminates the need for a batch layer, which requires additional processing and storage resources.
 - is a simplified version of the Lambda Architecture that enables real-time data processing and analysis without the need for batch processing



■ <https://www.chaosgenius.io/blog/kappa-architecture/>

75

BIG DATA ARCHITECTURE - COMPARISON

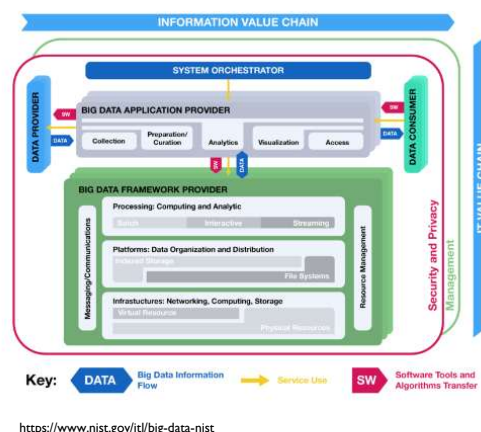
	Lambda Architecture	Arquitetura Kappa
Processing Model	Combines batch and real-time processing.	Concentra-se exclusivamente no processamento de fluxo (tempo real).
Layers	Batch tier, velocity tier, service tier.	Pipeline único para dados históricos e em tempo real.
Complexity	Increased due to separate batch and speed layer management.	Simpler as it uses only one layer of stream processing.
Fault tolerance	Fault-tolerant, as batch processing ensures data accuracy.	Fault-tolerant with real-time processing, but depends on the integrity of the stream.
Use Case	Suitable for real-time and batch processing needs.	Best for real-time processing; Batch processing is less emphasized.
Data reprocessing	Batch layer allows for accurate reprocessing of historical data.	Reprocessing is done by reproducing the stream in real time.
Latency	Increased latency in batch processing; Low latency in real time.	Low latency for all data due to stream processing.
Precision	The batch layer provides high precision, the speed offers immediate but less accurate results.	Provides consistent results, but may not match the accuracy of dedicated batch processing.

76

NIST BIG DATA REFERENCE ARCHITECTURE

- **Information Value axis**, the value is created through data collection, integration, analysis, and applying the results following the value chain.
- **IT axis**, the value is created through providing networking, infrastructure, platforms, application tools, and other IT services for hosting of and operating the Big Data in support of required data applications.
- The intersection of both axes is the **Big Data Application Provider** role, indicating that data analytics and its implementation provide the value to **Big Data stakeholders** in both value chains.

The National Institute of Standards and Technology (NIST)



77

NIST BIG DATA REFERENCE ARCHITECTURE

- The five main roles of the NIST Big Data Reference Architecture represent the logical components or roles of every Big Data environment:
 - **System Orchestrator** - is the automated arrangement, coordination, and management of computer systems, middleware, and services. Orchestration ensures that the different applications, data and infrastructure components of Big Data environments all work together. In order to accomplish this, the System Orchestrator makes use of workflows, automation and change management processes.
 - **Data Provider** - The Data Provider role introduces new data or information feeds into the Big Data system for discovery, access, and transformation by the Big Data system. The data can originate from different sources, such as human generated data (social media), sensory data (RFID tags) or third-party systems (bank transactions).
 - **Big Data Application Provider** - is the architecture component that contains the business logic and functionality that is necessary to transform the data into the desired results. The common objective of this component is to extract value from the input data, and it includes the following activities: Collection; Preparation; Analytics; Visualization; Access.
 - **Big Data Framework Provider** - has the resources and services that can be used by the Big Data Application Provider, and provides the core infrastructure of the Big Data Architecture. In this component, the data is stored and processed based on designs that are optimized for Big Data environments.
 - **The Big Data Framework Provider** - can be further sub-divided into the following sub-roles: Infrastructure: networking, computing and storage; Platforms: data organization and distribution; Processing: computing and analytic
 - **Data Consumer** - Similar to the Data Provider, the role of Data Consumer within the Big Data Reference Architecture can be an actual end user or another system. In many ways, this role is the mirror image of the Data Provider. The activities associated with the Data Consumer role include the following: Search and Retrieve; Download; Analyze Locally; Reporting; Visualization; Data to Use for Their Own Processes.

78

NIST BIG DATA REFERENCE ARCHITECTURE

- The benefits of using an open Big Data reference architecture include:
 - provides a common language for the various stakeholders;
 - encourages adherence to common standards, specifications, and patterns;
 - provides consistent methods for implementation of technology to solve similar problem sets;
 - illustrates and improves understanding of the various Big Data components, processes, and systems, in the context of a vendor- and technology-agnostic Big Data conceptual model;
 - facilitates analysis of candidate standards for interoperability, portability, reusability, and extensibility.

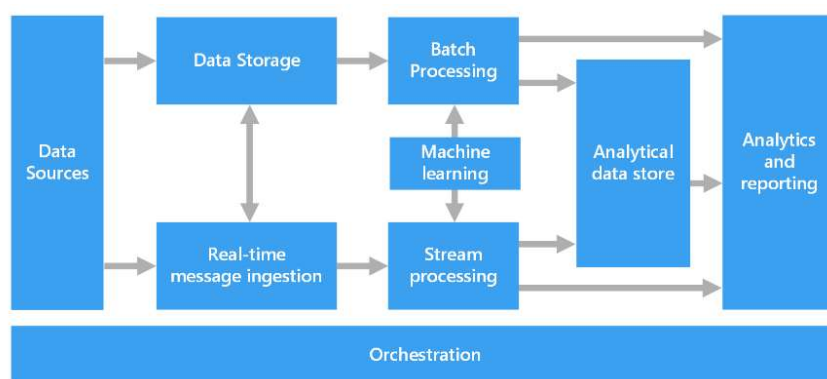
79

BENEFITS OF BIG DATA ARCHITECTURE

- **Parallel computing for high performance** - To process large data sets fast, big data architectures use parallel computing, in which multiprocessor servers perform numerous calculations at the same time. Sizable problems are broken up into smaller units which can be solved in parallel.
- **Elastic scalability** - Big Data architectures can be scaled horizontally, enabling the environment to be adjusted to the size of each workload. Big Data solutions are usually run in the cloud, where you only pay for the storage and computing resources you actually use.
- **Freedom of choice** - Many solutions and platforms for use in Big Data architectures, such as Azure managed services, MongoDB Atlas, and Apache technologies. Combine solutions to get the best fit for your various workloads, existing systems, and IT skill sets.
- **Interoperability with related systems** - Create integrated platforms across different types of workloads, leveraging Big Data architecture components for IoT processing and BI as well as analytics workflows.

80

BIG DATA ARCHITECTURE (AZURE)



Big data architectures - Azure Architecture - <https://learn.microsoft.com/en-us/azure/architecture/databases/guide/big-data-architectures>

81

COMPONENTS OF BIG DATA ARCHITECTURE

- **Data sources** – Data is sourced from multiple inputs in a variety of formats, including both structured and unstructured. Sources include relational databases allied with applications such as ERP or CRM, data warehouses, mobile devices, social media, email, and real-time streaming data inputs such as IoT devices. Data can be ingested in batch mode or in real-time.
- **Data storage** - This is the data receiving layer, which ingests data, stores it, and converts unstructured data into a format analytic tools can work with. Structured data is often stored in a relational database, while unstructured data can be housed in a NoSQL database such as MongoDB Atlas. A specialized distributed system like Hadoop Distributed File System (HDFS) is a good option for high-volume batch processed data in various formats.
- **Batch processing** - With very large data sets, long-running batch jobs are required to filter, combine, and generally render the data usable for analysis. Source files are typically read and processed, with the output written to new files. Hadoop is a common solution for this.
- **Real-time message ingestion** - This component focuses on categorizing the data for a smooth transition into the deeper layers of the environment. An architecture designed for real-time sources needs a mechanism to ingest and store real-time messages for stream processing. Messages can sometimes just be dropped into a folder; but in other cases, a message capture store is necessary for buffering and to enable scale-out processing, reliable delivery, and other queuing requirements.

82

COMPONENTS OF BIG DATA ARCHITECTURE

- **Stream processing** - Once captured, the real-time messages have to be filtered, aggregated, and otherwise prepared for analysis, after which they are written to an output sink. Options for this phase include Azure Stream Analytics, Apache Storm, and Apache Spark Streaming.
- **Analytical data store** - The processed data can now be presented in a structured format – such as a relational data warehouse – for querying by analytical tools, as is the case with traditional business intelligence (BI) platforms. Other alternatives for serving the data are low-latency NoSQL technologies or an interactive Hive database.
- **Analysis and reporting** - Most Big Data platforms are geared to extracting business insights from the stored data via analysis and reporting. This requires multiple tools. Structured data is relatively easy to handle, while more advanced and specialized techniques are required for unstructured data. Data scientists may undertake interactive data exploration using various notebooks and tool-sets. A data modeling layer might also be included in the architecture, which may also enable self-service BI using popular visualization and modeling techniques. Analytics results are sent to the reporting component, which replicates them to various output systems for human viewers, business processes, and applications. After visualization into reports or dashboards, the analytic results are used for data-driven business decision making.
- **Orchestration** - The cadence of Big Data analysis involves multiple data processing operations followed by data transformation, movement among sources and sinks, and loading of the prepared data into an analytical data store. These workflows can be automated with orchestration systems from Apache such as Oozie and Sqoop, or Azure Data Factory.

83

BIG DATA ARCHITECTURE CHALLENGES

- **Security** - Big data of the static variety is usually stored in a centralized data lake. Robust security is required to ensure your data stays protected from intrusion and theft. But secure access can be difficult to set up, as other applications need to consume the data as well.
- **Complexity** - A Big Data architecture typically contains many interlocking moving parts. These include multiple data sources with separate data-ingestion components and numerous cross-component configuration settings to optimize performance. Building, testing, and troubleshooting Big Data processes are challenges that take high levels of knowledge and skill.
- **Evolving technologies** - It's important to choose the right solutions and components to meet the business objectives of your Big Data initiatives. This can be daunting, as many Big Data technologies, practices, and standards are relatively new and still in a process of evolution. Core Hadoop components such as Hive and Pig have attained a level of stability, but other technologies and services remain immature and are likely to change over time.
- **Specialized skill sets** - Big Data APIs built on mainstream languages are gradually coming into use. Nevertheless, Big Data architectures and solutions do generally employ atypical, highly specialized languages and frameworks that impose a considerable learning curve for developers and data analysts alike.

84